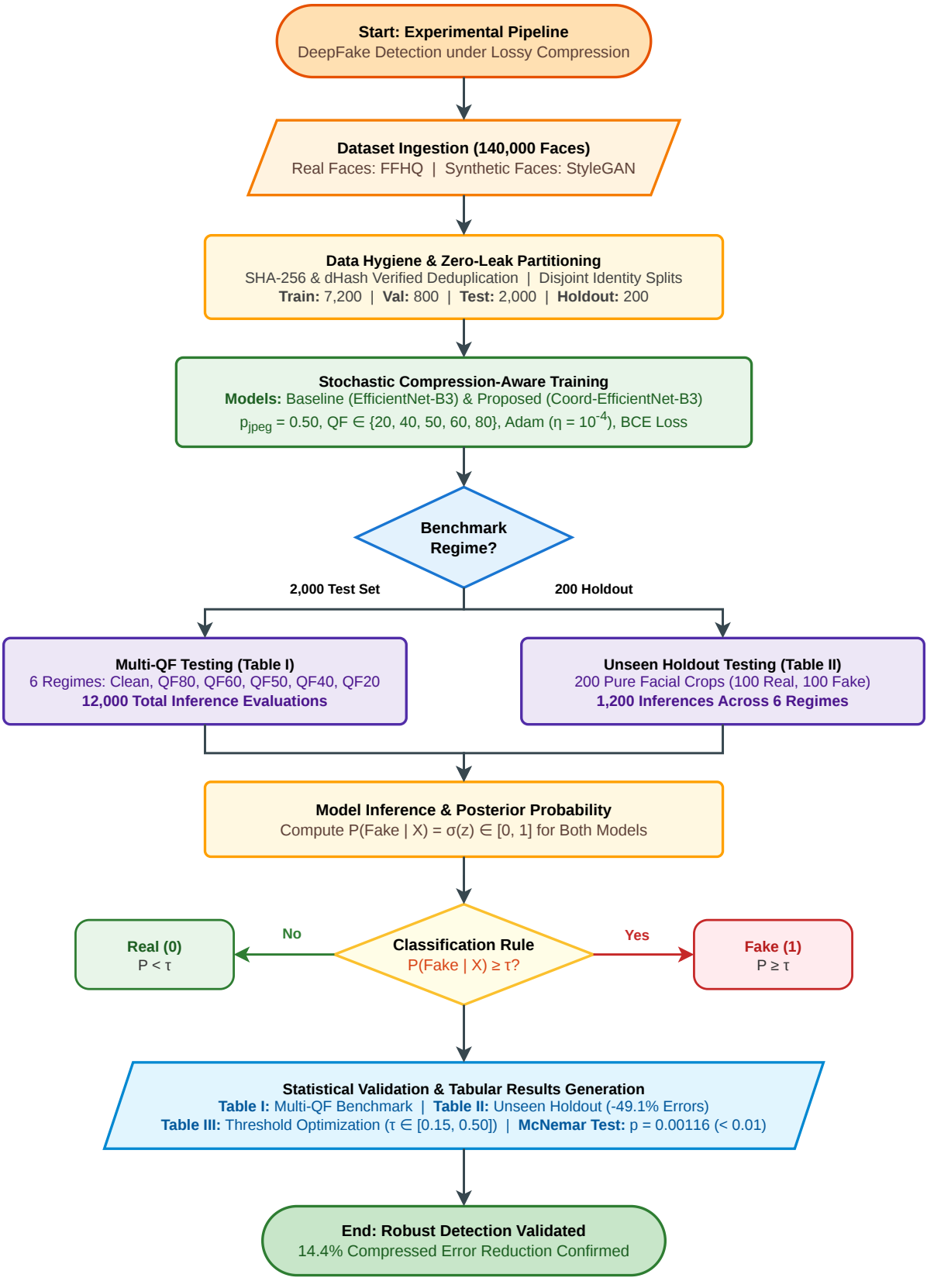
Start: Experimental Pipeline
DeepFake Detection under Lossy Compression
Dataset Ingestion (140,000 Faces)
Real Faces: FFHQ | Synthetic Faces: StyleGAN
Data Hygiene & Zero-Leak Partitioning
SHA-256 & dHash Verified Deduplication | Disjoint Identity Splits
Train: 7,200 | Val: 800 | Test: 2,000 | Holdout: 200
Stochastic Compression-Aware Training
Models: Baseline (EfficientNet-B3) & Proposed (Coord-EfficientNet-B3)
p_{jpeg} = 0.50, QF ∈ {20, 40, 50, 60, 80}, Adam ($\eta = 10^{-4}$), BCE Loss
Benchmark Regime?
2,000 Test Set
200 Holdout
Multi-QF Testing (Table I)
6 Regimes: Clean, QF80, QF60, QF50, QF40, QF20
12,000 Total Inference Evaluations
Unseen Holdout Testing (Table II)
200 Pure Facial Crops (100 Real, 100 Fake)
1,200 Inferences Across 6 Regimes
Model Inference & Posterior Probability
Compute P(Fake | X) = σ(z) ∈ [0, 1] for Both Models
Real (0)
P < τ
No
Classification Rule
P(Fake | X) ≥ τ?
Yes
Fake (1)
P ≥ τ
Statistical Validation & Tabular Results Generation
Table I: Multi-QF Benchmark | Table II: Unseen Holdout (-49.1% Errors)
Table III: Threshold Optimization (τ ∈ [0.15, 0.50]) | McNemar Test: p = 0.00116 (< 0.01)
End: Robust Detection Validated
14.4% Compressed Error Reduction Confirmed